Case Study: Apple - Health Data Collection through Apple Watch

David Umberger
Harrisburg University of Science & Technology

ANLY 620: Ethics for Data Analytics

Dr. Teray Johnson

December 4, 2025

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Executive Summary

The Apple Watch has developed as a major wearable gadget, providing continuous biometric and behavioral monitoring of heart rate, sleep cycles, physical activity, and oxygen levels. While these capabilities allow for individualized health insights and early disease identification, they also present substantial ethical, technological, and regulatory hurdles. This case study investigates Apple's health data collection procedures, focusing on concerns such as informed consent, privacy, algorithmic bias, and equitable access.

Apple's frameworks, HealthKit and ResearchKit, enable large-scale data gathering and exchange with healthcare providers, researchers, and insurers. Despite Apple's emphasis on privacy and security, the passive nature of data collecting and the complexities of consent agreements limit user autonomy. Unauthorized third-party access, monetization of sensitive health information, and potential prejudice when data is used in algorithmic decision-making are all risks to consider.

An examination of an Apple Watch dataset found structural integrity, but also outliers, skewed distributions, and imbalances in activity categories and gender representation. These biases increase the likelihood of misclassification, especially in health-related scenarios where correct detection is crucial. To address the imbalance, reweighting-based resampling was used, with a focus on justice and openness while avoiding synthetic data production.

Ethical judgment based on Kantianism and Utilitarianism highlights the tension between innovation and responsibility. Kantian ethics require truthful consent and equal treatment of users, whereas utilitarian approaches balance the life-saving possibilities of health monitoring against the hazards of privacy violations. Rule utilitarianism promotes adhering to data protection standards in order to maintain long-term trust.

The impacts of Apple Watch data harvesting are numerous. Improving fitness, monitoring heart health, detecting falls, and tracking medications are all positive effects. Privacy breaches, profiling concerns, psychological strain from "data fatigue," and regulatory gaps in consumer-grade health devices are some of the negative repercussions. Short-term benefits, such as emergency notifications, are offset by long-term concerns of algorithmic bias, misinterpretation, and the breakdown of doctor-patient relationships.

The recommendations include improving meaningful consent through progressive disclosure, bolstering security with end-to-end encryption and independent audits, enforcing stringent third-party data contracts, and releasing transparency reports. Apple should follow privacy-by-design principles, provide equitable access to all demographics, and keep explicit record of data cleaning and modeling procedures.

The ultimate challenge for Apple is to strike a balance between technological innovation and ethical responsibility. Transparent, safe, and fair health data policies are critical for maintaining user trust, protecting vulnerable groups, and ensuring that wearable technology contribute positively to public health while respecting human rights.

Introduction

The Apple Watch, a leading wearable health technology, collects a wide range of biometric and behavioral data, including heart rate, sleep cycles, physical activity, and blood oxygen levels. While this continuous, passive data collection enables personalized health monitoring and disease detection at early stages. It also introduces complex ethical challenges about user autonomy, the adequacy of informed consent, and the potential for data misuse and exploitation.

A study examined the "privacy self-management" model where users are expected to understand and manage their data privacy, which failed in the context of complex digital ecosystems, as individuals often lack the cognitive and informational resources to make fully informed decisions (Solove, 2013). Moreover, the integration of wearable data into broader health and insurance systems raises questions about data ownership, surveillance, and the commercialization of personal health information (Ajunwa, Crawford, & Schultz, 2017).

The ethical implications are further compounded by the risks of data breaches and unauthorized third-party access. As wearable devices become increasingly embedded in everyday life, the line between health monitoring and surveillance blurs, potentially undermining user trust and autonomy (Mittelstadt, 2017). The use of AI and machine learning to analyze wearable data introduces additional concerns, such as algorithmic bias and the opacity of predictive models, which may disproportionately affect marginalized populations (Sameh et al., 2025). These developments necessitate a critical examination of how companies like Apple can balance innovation with ethical responsibility, ensuring that health data collection practices are transparent, secure, and equitable.

Problem Statement

As wearable technologies become increasingly integrated into personal health monitoring, they collect massive amounts of biometric and behavioral data with minimal user intervention. While this data offers significant potential for early disease detection and personalized care, it also raises urgent ethical concerns. Users often lack the capacity to manage their privacy effectively within complex digital ecosystems, leading to inadequate informed consent and vulnerability to data misuse. This study aims to objectively examine the dataset and determine if the data is biased and unbalanced.

Research Questions

What are the ethical implications of Apple's health data collection through the Apple Watch?

How can the design and implementation of wearable technologies be improved to ensure meaningful user consent, robust data security, and protection against third-party misuse, while maintaining public trust and promoting equitable access to digital health innovations?

Facts Gathering

Apple introduced the Apple Watch in 2015, positioning it as both a lifestyle accessory and a health-monitoring device. Over sequential generations, the device has evolved to include advanced health features such as electrocardiogram (ECG) monitoring, fall detection, blood oxygen tracking, and irregular heart rhythm notifications. These capabilities have enabled Apple to position itself as a key player in the digital health system. They also have created alliances with healthcare providers, research institutions, and insurance companies.

Apple's HealthKit and ResearchKit frameworks further facilitate the aggregation and sharing of user health data for clinical and commercial purposes (Mittelstadt, 2017).

ResearchKit, an open-source software framework for medical research that enables investigators to create mobile applications, uses the iPhone's capacity to collect data, track movement, and take measurements. Even when Apple emphasizes privacy and user control in its public communications, the scale and sensitivity of the data collected raises relevant ethical concerns about how this information is used, stored, and potentially shared.

Since ResearchKit was released, Apple has developed alliances with healthcare providers, research institutions, and insurance companies. The primary stakeholders identified for this research are listed below.

- Consumers: Individuals who use the Apple Watch and whose health data is collected.
- Healthcare Providers and Researchers: Entities that may access aggregated or individual-level data for clinical or research purposes.
- Technology Companies: Apple and its partners who develop, manage, and monetize the data infrastructure.
- Regulators and Policymakers: Government bodies responsible for enforcing data protection laws and ethical standards.
- Third Parties: Including insurers, employers, and advertisers who may seek access to health data for risk profiling or marketing.

Ethical Analysis

Several ethical concerns emerge from Apple's health data collection practices. First, informed consent is often undermined by the complexity of data-sharing agreements and the passive nature of data collection. Users may not fully understand what data is being collected or how it might be used, which challenges the ethical principle of autonomy (Solove, 2013). Second, privacy and data security are critical concerns. Although Apple employs encryption and

on-device processing, no system is immune to breaches, and the aggregation of sensitive health data presents a high-value target for cyberattacks (Ajunwa, Crawford, & Schultz, 2017). Third, the potential for discrimination and bias arises when health data is used in algorithmic decision-making, such as in insurance underwriting or employment screening. These practices can disproportionately affect vulnerable populations, especially if the underlying data or algorithms reflect existing social biases (Sameh et al., 2025). Finally, the commercialization of health data raises questions about data ownership and the ethical limits of monetizing personal health information.

Ethical Frameworks

When examining ethical concerns, the following frameworks must be considered to provide a comprehensive analysis.

According to Kantianism, moral law and universal morals guide behavior through reason. Therefore, Apple should always be truthful about informed consent, as moral law dictates behaving in ways that respect others' rights. The company must view individuals as autonomous beings, recognizing their desires for data privacy and security. Users should have a say in how their data is used. Finally, within this framework, Apple should treat all customers equally, ensuring that algorithms and data distribution are fair, preventing disadvantages or discrimination against any group.

Act Utilitarianism suggests an action is beneficial if its advantages outweigh its drawbacks, based on the greatest happiness principle. For instance, Apple must evaluate if the health benefits of Apple-watch data collection surpass risks like data leakage and health data monetization, with the potential to save lives. In another scenario, should Apple breach HIPAA

rules for a few customers to alert a larger group about a virus outbreak? The benefits of timely alerts must be weighed against the negatives of data privacy violations.

Rule Utilitarianism, on the other hand, advocates for adhering to a moral rule that produces the greatest happiness. In the outbreak scenario, it argues against violating HIPAA for a broader warning, emphasizing the importance of trust from all customers. Apple should follow all data privacy regulations to maintain customer trust and ensure the protection of personal health information (PHI), promoting long-term benefits by following rules consistently.

Data and Algorithmic Issues

Data Source

The dataset used in this project was obtained from the Harvard Dataverse repository (Whelan, 2021). It originates from a controlled laboratory study designed to test whether commercial wearable devices can accurately predict different levels of physical activity. Forty-six participants (26 women) with a range of age from 18 to 56 wore an Apple Watch Series 2, Fitbit Charge HR2, and GENEActiv sensor during a 65-minute protocol that included treadmill walking, sitting, and lying tasks. Indirect calorimetry provided the reference energy-expenditure data.

The analytic dataset includes 3,656 minutes of Apple Watch observations, containing heart rate, step, distance, and calorie variables, along with derived metrics such as entropy, correlation, and normalized intensity. The dataset offers high-quality, minute-level information but represents a small, homogeneous sample of healthy adults under laboratory conditions.

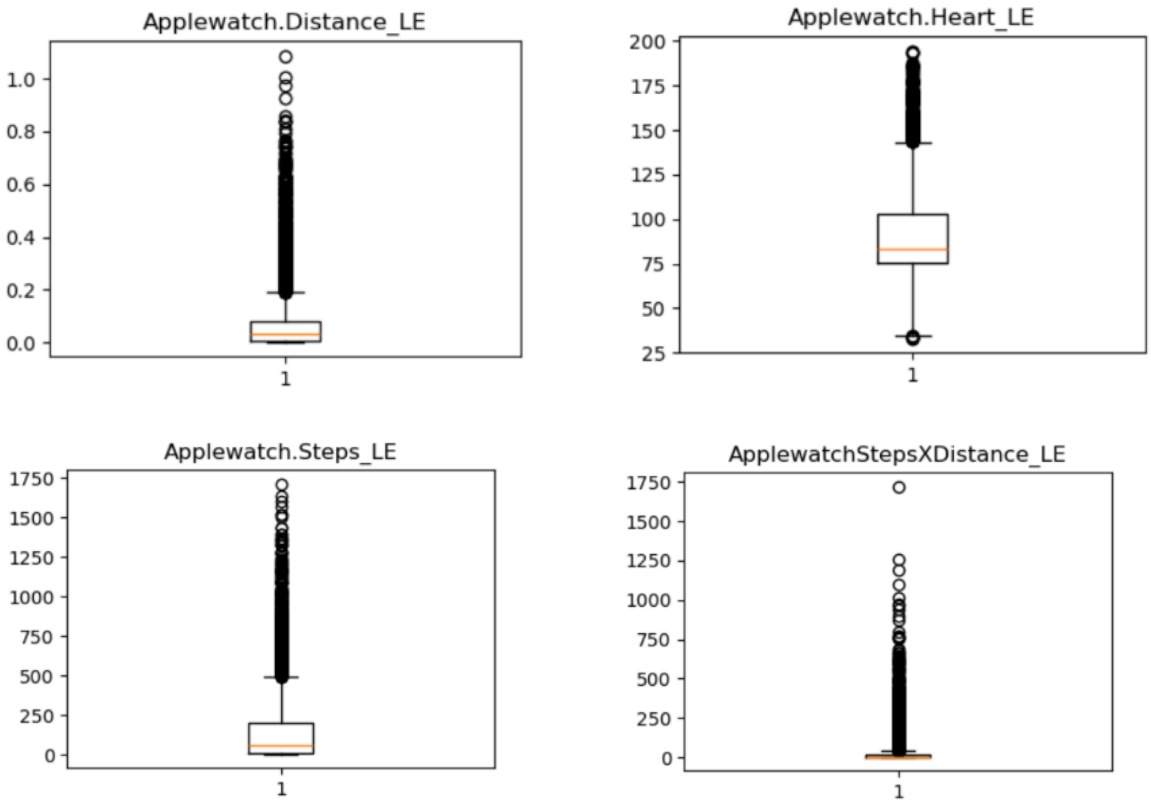
Data Quality Issues

Data quality evaluation began with a full structural and complete audit of the dataset. Using Python's isnull().sum() and duplicate checks, the analysis verified that no missing data or

duplicated records were present, confirming overall dataset integrity. This ensured that subsequent analyses were not affected by incomplete or redundant information.

Next, exploratory visualizations were created to assess distributional characteristics. Histograms and boxplots were generated for all Apple Watch numerical features, including heart rate, steps, distance, and calorie variables. The boxplots revealed that several features contained notable outliers, particularly steps and distance, where over 10 percent of observations fell outside the interquartile range. Such concentration of extremes suggests sensor spikes or activity-transition artifacts rather than random measurement noise.

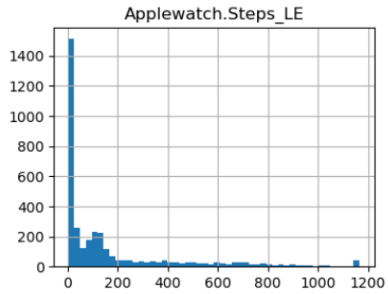
Graph 1 ***Variables that have more than 10% outliers***

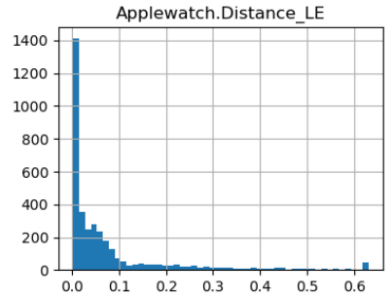


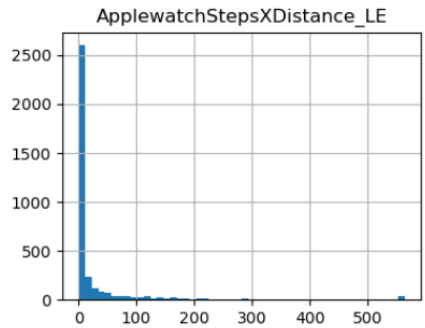
Graph 1 illustrates the variables with more than 10 percent outliers, providing a visual confirmation of which features required mitigation. To address these anomalies, winsorization (1st–99th percentile clipping) was applied to numerical variables that helped minimize the influence of extreme sensor spikes while preserving the true physiological range of movement. Post-cleaning descriptive statistics confirmed that all variables were brought within plausible ranges, enabling fair and stable model inputs and reducing the likelihood of bias introduced by erratic sensor readings.

Histogram on each variable was conducted in exploratory data analysis, which confirmed the Apple Watch's reliability in distinguishing activity levels, but also highlighted the need for transparency about internal smoothing algorithms. Over-smoothed data could suppress natural variability, which may disadvantage certain users such as women, older adults, or those with heart irregularities, whose physiological signals differ subtly.

Graph 2 ***Histograms of Steps, Distance, and Steps×Distance***

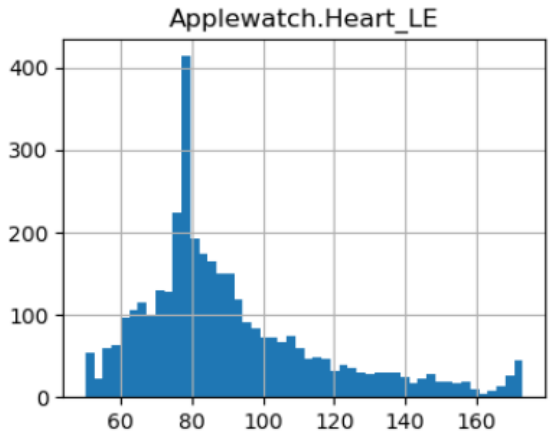


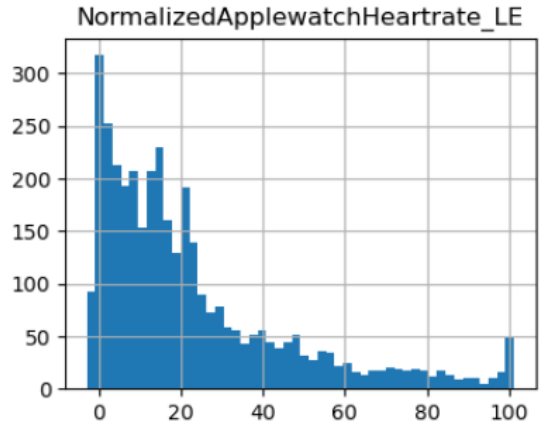




More specifically, Steps, Distance, and Step×Distance show right skewed distributions, meaning most activity minutes are low intensity, with a few high intensity spikes. This aligns with real world movement, where high exertion is brief, confirming that the earlier "outliers" (>10%) represent transitions or intense intervals, not random sensor noise.

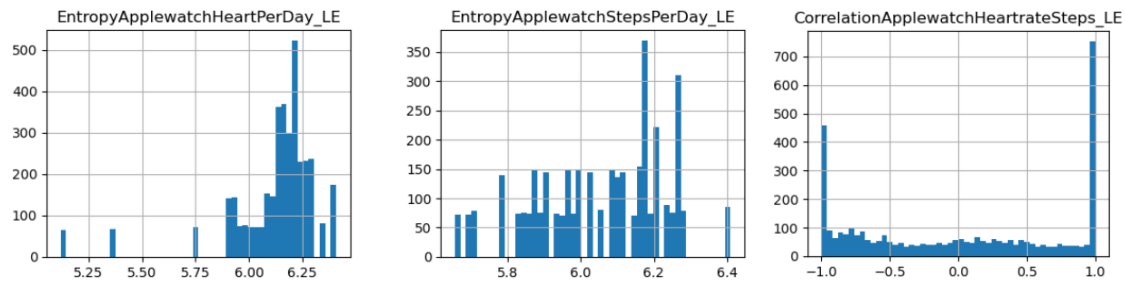
Graph 3 ***Heart Rate and Normalized Heart Rate***





Heart Rate and Normalized Heart Rate show bimodal patterns, which likely reflect the rest (sitting/lying) vs. exertion (walking/running) in terms of activities. This validates the activity labels and suggests that the Apple Watch sensors capture physiological variation accurately during protocol transitions.

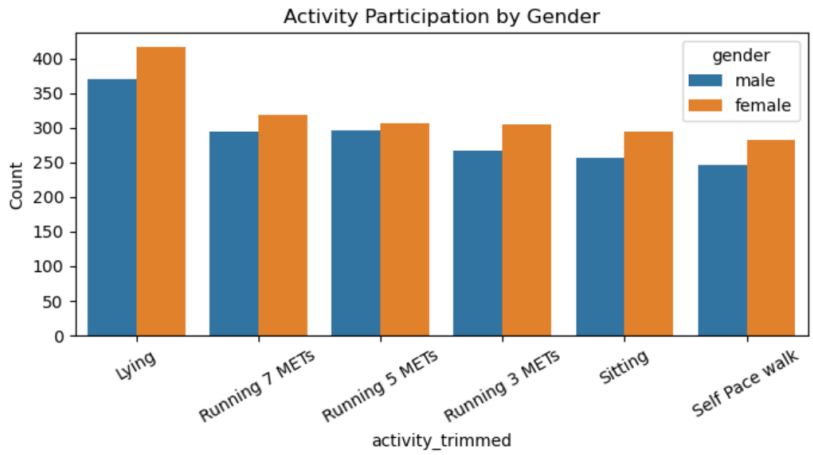
Graph 4 ***Entropy and Correction***

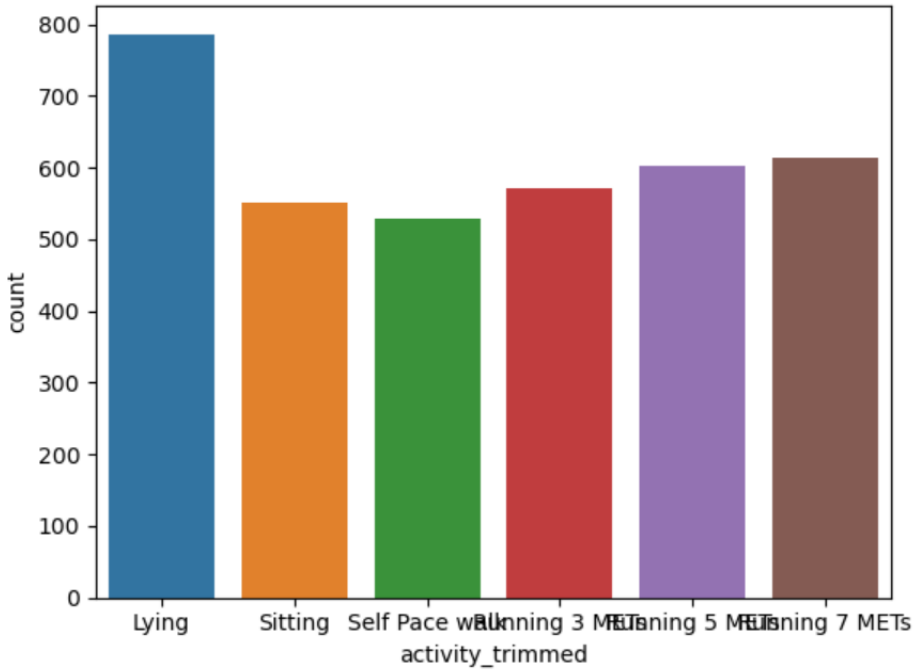


Entropy and Correlation metrics are narrow and stable, indicating that the device's internal algorithms already apply smoothing and denoising. This reinforces the idea that these derived variables carry less measurement noise but also might mask subtle individual differences.

Evaluation of Unbalanced Data

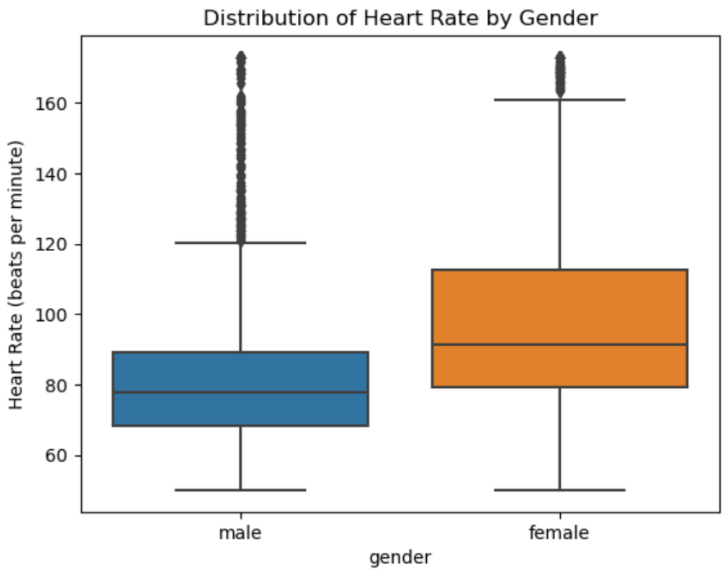
Graph 5 ***Activity Histograms***





The exploratory analysis revealed notable imbalances at both the activity-label level and the demographic level. The activity_trimmed count plot demonstrated that “Lying” overwhelmingly dominates the dataset across both genders, while the remaining activities, including Sitting, Walking, and Running at 3, 5, and 7 MET levels, are comparatively balanced and do not exhibit substantial skew. This pattern indicates label imbalance, wherein one class receives disproportionately more observations than others. In such cases, predictive models tend to favor the majority class because doing so minimizes overall error, even when misclassifying minority classes. This leads to representation bias that reduces model fairness and limits the reliability of predictions for less frequent activities.

Graph 6 ***Distribution of Heart Rate by Gender***



A second imbalance was observed in the gender distribution, with 26 women and 20 men represented in the sample. The cross tabulation of activity by gender showed that women appear more frequently in the “Lying” category, whereas men appear slightly more often in Walking or MET level activities. Although the differences are not extreme, they introduce an intersectional bias in which gender becomes indirectly associated with certain activity states. As a result, the model may implicitly learn that lower movement states correspond to female physiology, while higher-intensity activities correspond to male physiology. Given that physiological signals such as heart rate and step variability differ by sex, these imbalances heighten the risk of gender linked misclassification, particularly during borderline or transitional activity states.

Algorithmic Analysis

The dominance of the “Lying” class poses significant implications for algorithmic performance. When one activity type is substantially overrepresented, the model may systematically under-recognize active behaviors, resulting in underestimation of exercise duration, energy expenditure, or transitions between activity states. From an ethical standpoint,

this constitutes a form of data imbalance bias, which undermines fairness in contexts where equal performance across activity categories is necessary. This issue is particularly relevant in health-related applications, where false “resting” predictions may obscure early warning signs or inaccurately represent a user’s physical activity level.

The gender imbalance further contributes to potential representation bias, as the model may inadvertently encode gendered associations between heart rate patterns, step counts, and activity classifications. Such encoding may reinforce unwarranted stereotypes (e.g., women being more sedentary) or lead to physiologically inaccurate interpretations that disproportionately affect one gender. These risks underscore the importance of addressing demographic imbalance during model development and evaluation.

Improvement Techniques

To mitigate the identified imbalances, a reweighting-based resampling approach was implemented. This method applies inverse-frequency weights to each gender-activity combination, thereby up sampling underrepresented groups (e.g., male-Walking) and down sampling overrepresented groups (e.g., female-Lying). Reweighting was selected over methods such as SMOTE or SMOTE-ENN for three reasons: It avoids generating synthetic physiological signals, which is important for maintaining data authenticity in health-sensor contexts. It is interpretable and transparent, which supports accountability in healthcare-related modeling. It directly encodes fairness constraints by adjusting sampling probabilities according to demographic and activity distributions. This approach aligns with ethical principles of data integrity, transparency, and equitable model performance, ensuring that each subgroup contributes proportionally to the learning process.

Transparency Measures

Several steps were taken to enhance transparency throughout the analysis. First, feature source clarity was maintained by documenting whether each variable originated from raw sensor data (e.g., heart rate, steps, distance) or device-derived processing (e.g., entropy, correlation). Second, data quality documentation was provided by reporting missingness, identifying outliers, and presenting distribution plots that justify preprocessing decisions. Third, bias visibility was improved by reporting gender, activity, and intersectional distributions both before and after reweighting. Finally, model interpretability was emphasized through the use of human readable evaluation metrics such as accuracy and macro-F1.

To strengthen ethical accountability, the following additional steps are recommended: Publish a concise data-processing log detailing cleaning procedures, thresholds, and model parameters. Create a model card or dataset factsheet summarizing demographic composition, preprocessing steps, and bias-mitigation strategies. Maintain fully reproducible code, including the reweighting script, in appendix or supplementary materials.

Evaluation of Positive and Negative Impacts

Apple watch - Health Data Collection carries several positive and negative effects, which are described below.

Positive Impacts

- Physical activity and fitness – tracks activity and workouts goals stimulating and incentivizing activity and progress towards a healthy lifestyle
- Heart health – monitors heart rate including irregular rhythms
- Sleep tracking – watches stages, durations, apnea detection
- Mental and sensory health – follows emotions and daily moods, headphone levels

- Cycle tracking – for period and fertile window predictions (for women)
- Time in daylight – follows physical and mental health regarding exposure to sunlight
- Fall detection – alert and location of the user's physical fall
- Medication logging – tracks the user's vitamins/supplements & medicine

Negative Impacts

- Privacy Issues: Data such as heart rate, sleep habits, blood oxygen, and electrocardiogram readings are continuously gathered. Fitness tracker data frequently lacks the same legal protections as medical records covered by HIPAA. Users might not be completely aware of how their data is shared, stored, or made profitable.
- Risks to Data Security: Hackers can target even big businesses like Apple where location history, activity habits, and personal health measurements could all be compromised. Attackers may be able to access linked accounts and health information if login credentials are weak or repeated.
- Third-Party Distribution: Health data may be requested by apps linked to the Apple Watch and once shared, there is little control over the data, and businesses may use it for analytics or advertising. Aggregated health data may be used by employers or insurance firms to decide on workplace policies or premiums.
- Consent and Ethics Concerns: When enabling health tracking, many users fail to read or comprehend consent forms since they are written for a lawyer. This raises moral concerns about whether people actually consent to the level of data usage and surveillance.
- Monitoring and Profiling: Constant monitoring can build a digital profile of the user's lifestyle, including stress levels, exercise routines, and sleep patterns. This profile may be used for discriminatory or targeted marketing

- Psychological impact: Anxiety or a fixation with health indicators, commonly known as "data fatigue," might result from constant monitoring. Without medical context, users may misinterpret raw data, causing needless anxiety or false assurance.

Short-Term and Long-Term Effects

Overall, all devices can have positive and negative effects on final users. In this section, those effects are broken down into short and long-term effects as follows.

Positive Short-Term Effects

The short-term aspects considered in this study correspond to the immediate health benefits of the Apple watch and the data collected from it.

One important feature is the detection of a health issue that the user may be unaware of, perhaps saving one's life. If the watch senses a “fall”, it will give an alert, which the user can dismiss, but if the user did fall and is conscious and unable to respond, then a distress signal is sent for 911 paramedics for assistance. Otherwise, the customer may be in a critical situation, unable to call for help.

With physical activity and fitness features, Apple-watch can encourage exercise which has been shown to increase energy and life vitality. Heart monitoring could gather potential warning signs that would be largely undetected if one does not notice them.

Medication tracking could be a lifesaving short term feature because certain medications and steroid medicines cannot be missed or overdone, so any reminders' notifications are greatly helpful and critical relating to severity.

Negative Short-Term Effects

If someone were to gain access to a user's data and track their exercise route, they might potentially attempt to set a trap for them to rob them, or worse, physically harm them. Unfortunately, we have to consider those risks as someone could attempt such actions.

Positive Long-Term Effects

When certain health issues are not properly addressed in the short term, they could become long term problems that can be more serious. Apple-watch can help users by identifying patterns in areas like the ones listed below.

- Sleep tracking, involving sleep apnea, can become fatal since it involves sufficient air during sleep cycles.
- Mental health issues, which involve emotional balance, if not caught earlier, could continue to become deeper problems leading to worse issues, including suicide.
- Sensory health is longer in that prolonged exposure to high noise levels, including headphones, can result in hearing loss.

Negative Long-Term Effects

In this section, the effects below have been identified as potentially severe risks in the long term.

- Misinterpretation including false positive/negatives: When users get an incorrect diagnosed, they risk valuable time and resources trying to cure a disease that is non-existent while the real problem is not properly treated.
- Undermining Doctor-Patient relationships: High exposure to continuous "health" recommendations and good practices can lead users to think they are knowledgeable and can determine by themselves their disease and what they need to treat their condition.

- Overwhelmed users due to constant monitoring of biometrics: Having access to continuous notifications and alerts could have a negative effect on users' emotional well-being.
- Algorithm bias: The algorithms are typically developed with unbalanced and biased data sets that lead to incorrect results for users from demographics that were not considered in the training.
- Vulnerability to Breaches: The sensitive, intimate nature of health data (heart rate, sleep patterns, blood oxygen, activity levels) makes it a prime target for security breaches and potential identity theft. In one instance, over 61 million fitness tracker records from Apple and Fitbit users were exposed online.
- Third-Party Sharing: Users may be unaware that their data can be shared or sold to third-party entities like insurance companies or advertisers. This raises questions about who ultimately owns the data and how it is used for commercial purposes not directly related to the user's health.
- Regulatory Gaps: Data collected by consumer-grade smartwatches often falls into a regulatory "grey zone" and may not be protected by strict health information laws like HIPAA in the United States, as the devices are primarily classified as consumer goods, not medical devices

Ethical Frameworks to further explore the research question

When examining ethical concerns through the lenses of Kantianism and Utilitarianism, we can gain insights into ethical decision-making frameworks relevant to corporate behavior, particularly in the context of a technology company such as Apple.

Kantianism posits that moral law, grounded in universal ethics, directs human behavior. This ethical perspective emphasizes the role of reason in discerning right from wrong while providing justification for actions.

Within this framework, it is imperative that Apple upholds principles of honesty concerning informed consent. Even when such transparency might deter certain customers from opting in to data sharing, Kantian ethics asserts that moral law must guide actions—treat individuals with respect due to autonomous agents who have a right to be informed. It is crucial that the company communicates with its users in a manner consistent with how it would want to be treated.

Furthermore, Apple should treat everyone as an end to themselves rather than mere identifiers, such as iTunes account IDs. Recognizing customers as autonomous beings with specific needs and concerns regarding data privacy and security is essential, as they entrust their personal information to Apple for the purposes of health and well-being. Consequently, users must be able to afford a voice in the management of their personal data.

Lastly, the principle of equality necessitates that Apple treats all customers equitably, irrespective of their purchase history, device usage, or loyalty to the brand. The algorithms and underlying data practices should be uniformly applied to all users who utilize similar products for identical purposes. Ensuring an equitable distribution of data is paramount to preventing discrimination and bias from emerging in data processing.

In contrast, Act Utilitarianism assesses the morality of an action based on its consequences, specifically evaluating whether the benefits outweigh the drawbacks. This ethical framework is anchored in the "greatest happiness" principle, which measures the overall happiness or unhappiness derived from specific actions.

For instance, Apple must investigate whether the health benefits derived from the data collected through its health devices surpass the potential risks associated with data breaches, excessive data-sharing agreements, data biases, and the commercialization of health information. While opinions may diverge on this matter, a compelling argument exists that the potential health benefits, including life-saving interventions, warrant serious consideration.

A critical ethical dilemma arises concerning whether Apple should breach HIPAA regulations for a limited group of customers to alert a larger population at risk of a virus outbreak in a specific locale. This scenario requires careful balancing of the benefits of public health notifications against the ethical implications of compromising data privacy.

Similarly, Rule Utilitarianism advocates adherence to moral rules that, if generally followed, would yield the greatest happiness for all parties involved.

Returning to our virus outbreak scenario, Rule Utilitarianism suggests that Apple should refrain from violating HIPAA regulations for a small number of customers to alert a broader segment at risk. The long-term repercussions of maintaining customer trust outweigh the immediate benefits of alerting a larger audience, reinforcing the importance of abiding ethical standards.

Additionally, Apple must adhere to all data privacy regulations to preserve customer trust and safeguard Protected Health Information (PHI). Compliance with established rules encourages equitable treatment of data across all users, effectively mitigating bias and enhancing the integrity of the data management process. In short, adhering to the rule without exception is the moral thing to do.

Recommendations

Improving the Design and Implementation of Wearable Technologies

- Meaningful User Consent: Use progressive disclosure to show essential information upfront while allowing access to more detailed information as needed.
- Robust Data Security: Implement end-to-end encryption for all health data transfers, use multi-factor authentication, and regularly conduct independent security audits, publishing the results.
- Protection Against Third-Party Misuse: Require data usage contracts for third-party developers, including penalties for misuse, and introduce user alerts when third-party apps access sensitive data.
- Maintaining Public Trust: Adopt privacy-by-design principles: embed security and ethics into the system from the beginning rather than adding them later and publishing annual transparency reports detailing health data handling practices.
- Promoting Equitable Access: Ensure cross-platform compatibility so that health innovations are not limited to a single ecosystem. Design user interfaces that are accessible to older adults and people with disabilities

Ethical Recommendations for Apple's Health Data Collection

- Transparent and Informed Consent: Provide clear and simple explanations of the health data collected, how it is used, and who can access it.
- Data Minimization: Collect only the minimum necessary health data needed for functionality. Avoid storing sensitive biometric data longer than required.
- On-Device Processing: Prioritize local analysis of health metrics (e.g., interpreting ECG readings directly on the watch) to reduce exposure risks.

- Third-Party Restrictions: Implement strict vetting of third-party apps that request access to HealthKit data. Conduct audits and publish transparency reports on how third-party data is used.
- Equitable Access: Ensure that features are inclusive across diverse demographics, avoiding bias in algorithms (e.g., calibrating heart rate sensors for various skin tones). Provide affordable options or subsidies to prevent inequalities in digital health access.

Regulation & Legal Framework

- Center civil society: Given the potential for harm, it is imperative to aim for balance between civil society and directly impacted communities so research centers consider their input to identify unintended harm and ensure regulations that protect human rights.
- Regulate corporate power: Antitrust laws are needed to help curb the dominance of major tech companies. Limiting the market power of these companies can reduce their ability to dictate the terms of tech development and adoption.
- Incentivize ethical behavior: Governments can create incentives for companies that go above and beyond legal requirements to ensure ethical AI development. This could include preferential treatment in government contracts or public recognition.
- New regulations: Create regulations at the Federal level to prevent data transfers to organizations with conflicted interests such as insurance companies.

Conclusions

- In 2015 Apple introduced the Apple-watch and ResearchKit which is an open-source software introduced for medical research, that enables investigators to create mobile applications.

- As of today, the device encompasses advanced health features, and Apple has developed alliances with healthcare providers, research institutions, and insurance companies.
- Health care systems, insurance, workplace dynamics, and social fairness are all impacted by the gathering of health data via wearable technology, such as the Apple Watch.
- Without strong protection, the benefits of these technologies could be overshadowed by privacy concerns, discrimination, and unequal access to health resources.
- To maintain public trust, wearable technologies must prioritize privacy, equitable access, and transparency. Ignoring these principles may worsen existing inequalities and normalize surveillance in everyday life.

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